Emily Tan

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Education

University of California, Riverside — Riverside, CA
Bachelor of Science in Computer Science (Honors) — *Expected 2029*

Experience

Cutie Hack Lead Software Engineer | Cutie Hack Github — Apr. 2026 – Present
Association for Computing Machinery @UCR — *Riverside, CA*

- Building and debugging software for Cutie Hack, an ACM hackathon serving 300+ participants
- Coordinate with software and UI/UX leads to establish deadlines and track development progress
- Reviewed pull requests and resolved GitHub issues

ACM Spark Web Developer | Rides | SHPE | AEP — Jan. 2026 – Aug. 2026
Association for Computing Machinery @UCR — *Riverside, CA*

- Collaborated in an Agile, GitHub issue-based workflow to deliver weekly frontend features from Figma designs
- Developed scalable UI components using Next.js, TypeScript, and Tailwind CSS for production deployment
- Implemented responsive design patterns to ensure consistent performance across devices and screen sizes

ACM Forge Fullstack Developer | GitHub — Apr. 2026 – Jun. 2026
Association for Computing Machinery @UCR — *Riverside, CA*

- Collaborated in a multidisciplinary team to develop a smart surveillance system with real-time video detection
- Focused on full-stack development using Next.js, React, and TypeScript to build live monitoring dashboard
- Integrated computer vision streaming pipelines into the frontend using Flask video feeds and MQTT messaging

Computer Science Associate — Oct. 2025 – Present
Formula SAE Highlander Racing @UCR — *Riverside, CA*

- Developing a front of car test bench to test vehicle control logic
- Researched redesigning control logic using state machines and ADC DMA to reduce CPU load
- Programmed an STM32 microcontroller to transmit custom CAN bus messages and verified their accuracy using an oscilloscope and CAN Bus transceiver module

Projects

UCR Registration Extension | Webstore Link — GitHub

- Built a Chrome extension that aggregates course data from multiple sources into a unified registration interface
- Integrated MyClassGrades, UCR course difficulty data, and RateMyProfessors into a structured data pipeline
- Implemented SPA-aware DOM injection with service workers and Chrome storage APIs for persistent performance
- Deployed the extension, helping 100+ students determine class difficulty during peak registration season

Mount Cryptid | *C++, UML, GoogleTest, CI* — GitHub

- Designed UML diagrams to establish game architecture and class relationships
- Collaborated in a 4-person team using an Agile Git workflow with feature branches and pull requests
- Developed and tested game features with GoogleTest and CI to maintain code quality and catch regressions

Pill-E – BioEngineering Makeathon — GitHub

- Developed control logic for precise pill dispensing and real-time device state management
- Implemented device-to-mobile communication layer to interface with a companion app built by teammates
- Debugged and validated hardware performance through iterative testing and calibration of the dispensing system

Technical Skills

Languages: C++, Python, TypeScript, JavaScript
Web Development: React, Next.js, Node.js, HTML/CSS, Tailwind CSS
Embedded: STM32, CAN Bus, Arduino, PlatformIO, MQTT
AI/ML: YOLO, scikit-learn
Data Science: pandas, NumPy, Matplotlib